

Problem 1: 2D Heat Diffusion with Convergence Detection

Consider an $N \times N$ temperature grid representing a two-dimensional material. At every iteration, update the temperature of each interior cell using the five-point stencil:

$$T(i,j)^{t+1} = [T(i-1,j)^t + T(i+1,j)^t + T(i,j-1)^t + T(i,j+1)^t] / 4$$

The simulation must continue until the maximum absolute change between two consecutive grids becomes smaller than a specified tolerance ϵ .

Develop a CUDA-based implementation that performs the heat-diffusion simulation on the GPU. The implementation should compute the convergence criterion in parallel and should avoid unnecessary transfers of the complete temperature grid between the CPU and GPU.

The program should support **fixed-temperature boundary conditions**. Compare a straightforward global-memory implementation with an implementation that uses shared-memory tiling to reduce global-memory accesses.

Input: N , initial temperature grid, boundary conditions, and ϵ . Output: the converged temperature grid, number of iterations, and execution time.